

Pranav Dhakal

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EDUCATION

Northern Kentucky University | Highland Heights, KY |

Jan 2021–Dec 2025

Bachelor of Science in Computer Science | Minor in Computer Information Technology | Graduated |

Relevant Coursework: Artificial Intelligence ,Data Structure and Algorithms, Object-Oriented Programming, Computer Systems, Introduction to Networking, Web Development, Block-Chain, Android App Development, Python, Raspberry Pi, Programming, Database Programming, Advanced Programming Method, Operating System, Design and Analysis of Algorithms, Server-Side Programming, Theory of Computation, Advanced Mathematics (Calculus ,Stats, Discrete Math)

SKILLS

Languages: Java, Python, C++, C#, SQL, React.js, JavaScript, Kotlin, Dart, SQL, HTML/CSS, .NET, PHP, Solidity, IT

Technology/Dev-tools: Flutter SDK, Android SDK, Google Firebase, Git, Cloud, Basic, Microsoft 365, Flask, GCP, API, State Management, Linux, IntelliJ, Android Studio, Visual Studio, Jira, Docker, Fast API

RESEARCH EXPERIENCE

Research Project: Scalable Long-Context Language Models | Py-Torch, Mamba (SSM)| Northern Kentucky University |

June 2025

Worked on research focused on addressing Transformer context-window limitations by exploring Mamba (State Space Models) for scalable long-context language modeling. This contributed to fine-tuning and evaluating a 7B-parameter language model using Py-Torch on NVIDIA A100 GPUs and comparing long-context performance against Transformer-based baselines. The work emphasized architectural tradeoffs, memory efficiency, and practical challenges in training large models on extended sequences. The work emphasized architectural tradeoffs, memory efficiency, and practical challenges in training large models on extended sequences.

Research Assistant at College of Informatics| Northern Kentucky University |

Aug 2023

Secure Hierarchical Edge Service Framework for IoT Devices

This research explores the creation of a hierarchical secure IoT-based edge service framework with advanced features such as an Intrusion Detection System and provenance data storage using blockchain technology. The proposed system adopts Edge-X Foundry, an open-source platform for IoT device management. Multiple IoT devices are connected to an Edge-X Server, which receives data from the nodes within the cluster.

WORK EXPERIENCE

Plus, Tutor for Java Programming | Northern Kentucky University

Aug 2023– 2024

Monitor and assess students' progress throughout the semester, offering constructive feedback and additional resources to support their academic success. Engaged with incoming freshmen to identify their academic and personal goals, providing a supportive environment for discussing concerns and aspirations Implemented online appointment scheduling systems, resulting in a 20% increase in appointment bookings and a 15% reduction in scheduling errors.

Volunteer Intern| Fellowship Society Nepal

Feb 2020 – Sept 2020

Administration duties, including data entry, documentation, and communication, collaborating remotely with the organization from, Nepal.

Programming Intern | Aarati Multi-Purpose Co-operative | Nepal

Oct 2019 - June 2020

Teamed up with the Full Stack Developer to optimize backend software, integrating modifications and unveiling new features to streamline account and data management, ultimately leading to a 25% reduction in processing time. Crafted and delivered a comprehensive design proposal for SMS implementation, yielding a 30% surge in customer engagement and communication efficiency.

PROJECTS

Machine Learning Model |Python|

Built and trained a machine learning model for predicting stock price movements using historical data. Implemented data preprocessing, feature engineering, and model evaluation.

Blockchain App| Solidity,JavaScript,React.js|

Led the design of Solidity smart contracts, conducting thorough testing with Node.js and JavaScript, and ensuring secure deployment via Web3 and Ganache CLI

Stock Insights – AI Stock Analysis App | Android Studio , Kotlin, Django, ML|

Built an Android app for real-time stock tracking and 7-day price forecasting using Facebook Prophet. Integrated Finn hub & Polygon APIs implemented Buy/Sell/Hold recommendations, and token-based authentication.

Raspberry Pi Laser Tripwire Security System | Python, Raspberry Pi|

Built a Raspberry Pi-based security system using a laser tripwire to detect intrusions, trigger alarms, capture images, and send email alerts.

Implemented reliable GPIO signal handling with a photodiode and transistor circuit and enabled remote arming and disarming via SSH on Linux.

Dorm Deal – Campus Marketplace | Next.js, TypeScript|

Developed a full-stack marketplace with listings, search, messaging, dashboards, and admin panel. Designed scalable architecture with Next.js App Router, Tailwind, and Firebase-ready services.

AI-Text-Detector | Python, JavaScript, Flask, JSON|

Developed a web application using GPT-zero API to identify AI-generated text, managing JSON with the request library. Technologies used: CSS, HTML, Flask framework, Python, JS

Algorithm Solver |Java|

Developed a program capable of solving various algorithms, including sorting, searching, and graph algorithms and demonstrated a deep understanding of algorithm design.

Data Visualization Dashboard |Python, Matplotlib|

Designed a data visualization dashboard for exploring and visualizing COVID-19 data. • Utilized Matplotlib for creating interactive visualizations

ACTIVITIES AND ACHIEVEMENTS

Awards: Edge Award, International Merit, International Diversity Scholarship

Honors: Dean's List (2022, 2023)